

Orthogonal Reparameterization of the Nelson-Siegel-Svensson Interest Rate Model

*A Mathematical Framework for Yield Curve Anomaly Detection
via Economically Interpretable Parameter Clustering*

Abstract. The Nelson-Siegel-Svensson (NSS) model is the workhorse of sovereign yield curve fitting, yet its six-parameter representation suffers from severe collinearity — particularly between the curvature and second-hump loadings — making raw-parameter monitoring unreliable for detecting stripping anomalies or data errors. We develop an orthogonal reparameterization that decomposes the NSS parameter vector into economically interpretable, statistically independent factors: level, slope, belly curvature, and hump location. By clustering daily yield curves in this orthogonal space, we construct a regime-aware anomaly detector that flags curves whose parameter signature is inconsistent with current market conditions — a symptom of bond stripping artefacts, quote errors, or structural breaks — without relying on RMSE, which is insensitive to the very distortions it should catch.

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1. The Nelson-Siegel-Svensson Model

The Nelson-Siegel (1987) framework models the instantaneous forward rate as an exponential polynomial. Svensson (1994) extended it with a fourth factor to capture a second hump, yielding the six-parameter specification adopted by the ECB, Bundesbank, BIS, and most sovereign debt offices for daily curve publication.

1.1 Functional Form

The zero-coupon continuously compounded yield at maturity m is given by:

$$y(m) = \beta_1 + \beta_2 \left(\frac{1 - e^{-m/\tau_1}}{m/\tau_1} \right) + \beta_3 \left(\frac{1 - e^{-m/\tau_1}}{m/\tau_1} - e^{-m/\tau_1} \right) + \beta_4 \left(\frac{1 - e^{-m/\tau_2}}{m/\tau_2} - e^{-m/\tau_2} \right)$$

We denote the maturity-dependent loadings (basis functions) as:

$$\phi_1(m) = 1, \quad \phi_2(m; \lambda_1) = \frac{1 - e^{-\lambda_1 m}}{\lambda_1 m}$$

$$\phi_3(m; \lambda_1) = \frac{1 - e^{-\lambda_1 m}}{\lambda_1 m} - e^{-\lambda_1 m}, \quad \phi_4(m; \lambda_2) = \frac{1 - e^{-\lambda_2 m}}{\lambda_2 m} - e^{-\lambda_2 m}$$

where we use the decay-rate parameterization:

$$\lambda_k = 1/\tau_k, \quad k \in \{1, 2\}$$

so that the compact vectorial form reads:

$$y(m; \theta) = \vec{\phi}(m; \lambda)^T \vec{\beta}, \quad \theta = (\beta_1, \beta_2, \beta_3, \beta_4, \lambda_1, \lambda_2)^T \in \mathbb{R}^6$$

1.2 Basis Functions

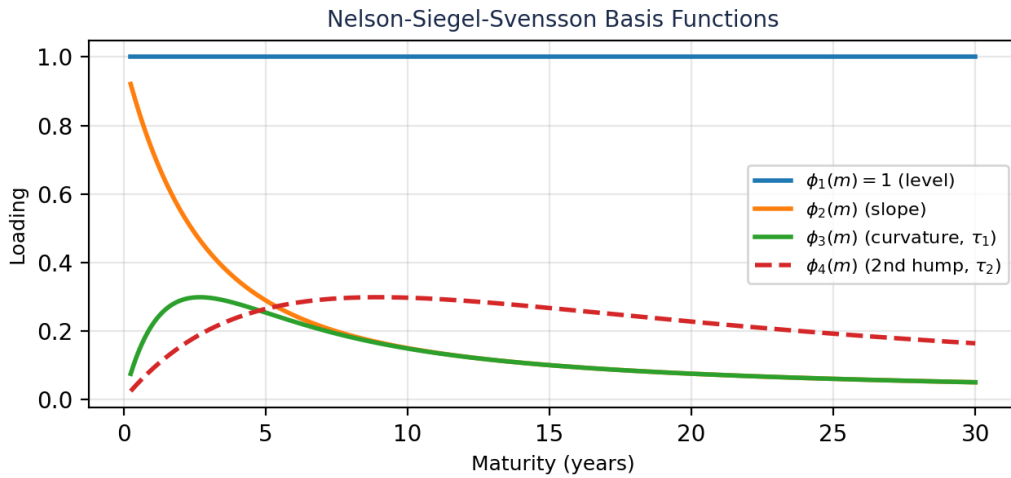


Figure 1. NSS basis functions for characteristic decay parameters. Note the substantial overlap between the curvature and second-hump loadings when the decay parameters are close.

1.3 Fitting Objective

Given N observed yields at maturities on date t , the fitting problem is the nonlinear least-squares programme:

$$\hat{\theta}_t = \arg \min_{\theta} \sum_{i=1}^N [y_i^{obs} - y(m_i; \theta)]^2 = \arg \min_{\theta} \|\vec{y}^{obs} - \Phi(\lambda)\vec{\beta}\|^2$$

where the design matrix has rows formed by evaluating the four basis functions at each observed maturity. For fixed decay parameters, the problem is linear in the betas and the conditional OLS estimator is:

$$\hat{\vec{\beta}}(\lambda) = [\Phi(\lambda)^T \Phi(\lambda)]^{-1} \Phi(\lambda)^T \vec{y}^{obs}$$

The full nonlinear optimisation reduces to a two-dimensional concentrated problem over the decay rates only, tractable but plagued by multiple local minima and a nearly flat objective surface — the root cause of the identification pathology discussed next.

2. Identification Pathology: Why Raw Parameters Mislead

2.1 Near-Collinearity of Basis Functions

The core problem is geometric: the columns of the design matrix are highly correlated whenever the two decay parameters are not well separated. The Gram matrix becomes ill-conditioned:

$$\kappa(\Phi^T \Phi) = \frac{\sigma_{\max}(\Phi)}{\sigma_{\min}(\Phi)} \gg 1 \quad \text{when } |\lambda_1 - \lambda_2| \text{ is small}$$

This means that a given yield curve can be fit almost equally well by many different combinations of the beta coefficients. The parameter vector lives on a near-degenerate manifold in parameter space:

$$\mathcal{M}_\varepsilon = \{\theta \in \Theta : \|\vec{y}^{obs} - y(\cdot; \theta)\| < \varepsilon\}$$

and this manifold is elongated along directions corresponding to large parameter swings with negligible effect on RMSE.

2.2 Concentrated Objective Surface

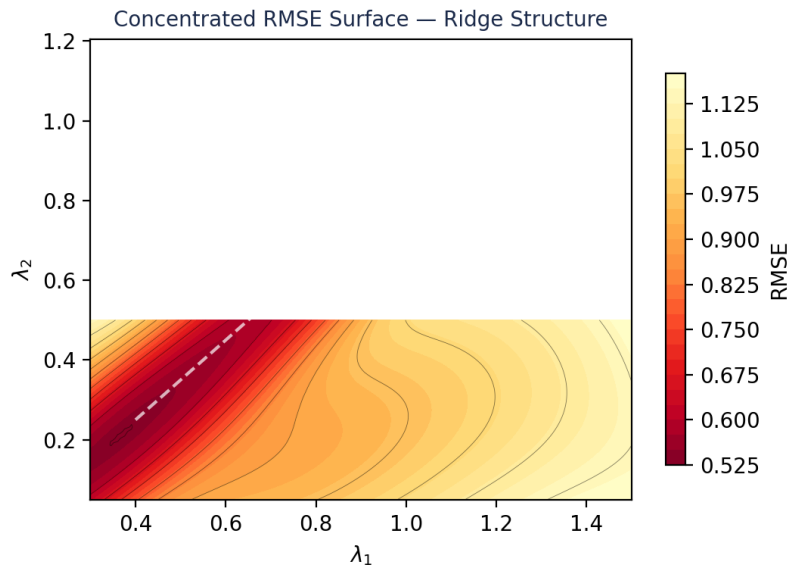


Figure 2. The concentrated RMSE surface over the two-dimensional decay-rate space exhibits a prominent ridge along the near-degenerate direction, explaining why different starting values lead to very different parameter estimates.

2.3 Empirical Correlation Structure

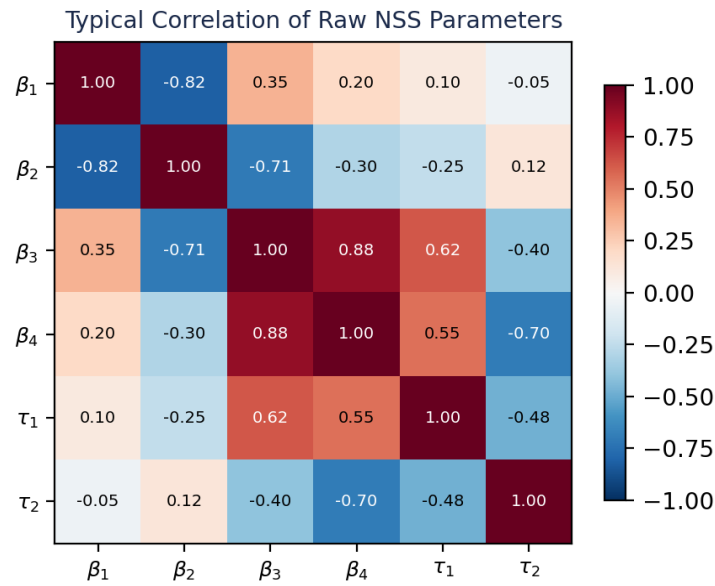


Figure 3. Typical pairwise correlation of daily NSS parameters estimated on sovereign curves. Note the extreme correlation between the curvature and second-hump betas.

2.4 Consequences for Anomaly Detection

Suppose we monitor the daily RMSE to flag stripping anomalies. A stripping error introduces a localised distortion — say a 5 bp dislocation at the 7-year point. The NSS model absorbs this distortion by jointly adjusting the curvature and hump parameters, producing a low RMSE but an economically implausible parameter vector:

- RMSE remains within tolerance (false negative for the anomaly detector).
- The parameter shift occurs along the near-degenerate direction, so individual parameter time-series exhibit jumps that are *artefacts of the parameterization* rather than genuine market moves.
- Conversely, genuine regime changes (e.g., a curve inversion) produce large RMSE spikes that are not anomalies at all.

This motivates a rotation into a space where each coordinate has a stable, unique economic meaning.

3. Orthogonal Reparameterization

3.1 Fixing the Decay Parameters

Following Diebold and Li (2006), we fix the decay parameters at values that maximize the average goodness of fit over a training window:

$$(\lambda_1^*, \lambda_2^*) = \arg \min_{\lambda_1, \lambda_2} \frac{1}{T} \sum_{t=1}^T \|\vec{y}_t^{obs} - \Phi(\lambda) \hat{\vec{\beta}}_t(\lambda)\|^2$$

subject to the ordering constraint:

$$\lambda_1 > \lambda_2 > 0$$

This constraint breaks the permutation symmetry between the two humps. With decay rates fixed, daily estimation becomes OLS and the beta vector inherits full identifiability.

3.2 Spectral Decomposition of the Beta Covariance

Over a rolling window of T trading days, collect the estimated beta vectors and compute the sample covariance:

$$B = (\hat{\vec{\beta}}_1, \dots, \hat{\vec{\beta}}_T)^T \in \mathbb{R}^{T \times 4}$$

$$\Sigma_\beta = \frac{1}{T-1} (B - \bar{B})^T (B - \bar{B})$$

The eigendecomposition provides the orthogonal directions:

$$\Sigma_\beta = V \Lambda V^T, \quad V = (v_1 \ v_2 \ v_3 \ v_4), \quad \Lambda = \text{diag}(\sigma_1^2, \sigma_2^2, \sigma_3^2, \sigma_4^2)$$

with eigenvalues ordered:

$$\sigma_1^2 \geq \sigma_2^2 \geq \sigma_3^2 \geq \sigma_4^2 > 0$$

3.3 The Orthogonal Factor Representation

Define the orthogonal beta vector as the rotation:

$$\tilde{\vec{\beta}}_t = V^T (\hat{\vec{\beta}}_t - \bar{\vec{\beta}})$$

By construction, the components are uncorrelated:

$$\text{Cov}(\tilde{\beta}_{t,j}, \tilde{\beta}_{t,k}) = \sigma_j^2 \delta_{jk}$$

and the yield curve can be written in terms of orthogonal factors:

$$y(m; \vec{\beta}_t) = \bar{y}(m) + \sum_{j=1}^4 \tilde{\beta}_{t,j} \psi_j(m)$$

where the *orthogonal basis functions* are:

$$\psi_j(m) = \sum_{k=1}^4 V_{kj} \phi_k(m; \lambda^*)$$

3.4 Relation to Principal Component Analysis

This construction is equivalent to performing PCA on the NSS beta vectors rather than on the yield levels directly. PCA on yields is model-free but loses the parametric structure; PCA on the NSS betas preserves the functional form while decorrelating the latent factors. The orthogonal factors inherit the smoothness prior implicit in the NSS specification.

The relationship between yield PCs and our orthogonal factors is:

$$\text{PC}_j^{\text{yield}}(m) = \psi_j(m) + \mathcal{O}(\varepsilon)$$

where the residual vanishes when the NSS fit is exact. This bridge provides the economic interpretation detailed next.

4. Economic Interpretation of the Orthogonal Factors

The rotated factors, while formally statistical objects, admit clean economic readings that extend the classical level-slope-curvature decomposition.

4.1 First Factor: Level

The first eigenvector is dominated by loadings on the constant and slope, capturing the parallel shift:

$$\tilde{\beta}_{t,1} \approx \alpha_1 \beta_{1,t} + \alpha_2 \beta_{2,t}, \quad |\alpha_1| \gg |\alpha_3|, |\alpha_4|$$

This factor is the dominant source of yield variation (typically 85-95% of total variance). Economically, it reflects aggregate term premium and long-run inflation expectations, driven by monetary policy, fiscal outlook, and global risk appetite.

4.2 Second Factor: Slope

The second eigenvector captures the differential between short and long ends:

$$\tilde{\beta}_{t,2} \propto y(m_{long}) - y(m_{short})$$

This maps to the classical 2s10s spread and carries information about the expected path of short rates. It is sensitive to business cycle positioning and central bank forward guidance.

4.3 Third Factor: Belly Curvature

The third factor captures richness or cheapness of the belly:

$$\tilde{\beta}_{t,3} \propto 2y(m_{mid}) - y(m_{short}) - y(m_{long})$$

This is proportional to the butterfly spread. Stripping anomalies are most visible here: a bond-strip mismatch at the 5-7 year point inflates this spread without a macro driver.

4.4 Fourth Factor: Hump Asymmetry

The smallest factor captures residual shape variation — the *difference* between the two NSS humps:

$$\tilde{\beta}_{t,4} \propto \beta_{3,t} \cdot g(\lambda_1) - \beta_{4,t} \cdot g(\lambda_2)$$

where g is a normalising function. This factor is the most diagnostic: it has low variance under normal conditions but spikes when the fit exploits the near-collinear direction to accommodate a data error.

4.5 Summary: Factor-Economic Mapping

Factor	Dominant NSS Components	Economic Proxy	Variance Share	Anomaly Sensitivity
Level (1st)	$\beta_{\blacksquare}, \beta_{\blacksquare}$	Long-run inflation expectations + premium	85–95%	Low
Slope (2nd)	$\beta_{\blacksquare}$	2s10s spread, policy stance	3–8%	Low

Belly Curvature (3rd)	$\beta_{\blacksquare}$	Butterfly spread, sector rich/cheap	1–4%	Medium
Hump Asymmetry (4th)	$\beta_{\blacksquare}, \beta_{\blacksquare}$ (difference)	Residual shape; strip/quote errors	<1%	High

Table 1. Mapping orthogonal factors to economic quantities and anomaly sensitivity.

5. Clustering Yield Curves in the Orthogonal Space

5.1 Feature Vector Construction

For each date t , the orthogonal transformation produces a standardised feature vector:

$$z_t = \Lambda^{-1/2} \tilde{\beta}_t = \Lambda^{-1/2} V^T (\hat{\beta}_t - \bar{\beta}) \in \mathbb{R}^4$$

The standardisation ensures each component has unit variance, preventing the dominant level factor from swamping the clustering metric.

5.2 Distance Metric

In the standardised space, the Mahalanobis distance simplifies to Euclidean:

$$d(z_t, z_s) = \|z_t - z_s\|_2 = \sqrt{\sum_{j=1}^4 \frac{(\tilde{\beta}_{t,j} - \tilde{\beta}_{s,j})^2}{\sigma_j^2}}$$

This weights deviations in the low-variance hump factor as heavily as those in the high-variance level factor — exactly matching our priorities.

5.3 Clustering Algorithms

Gaussian Mixture Models (GMM). Each cluster represents a yield-curve regime. The generative model is:

$$z_t \sim \sum_{k=1}^K \pi_k \mathcal{N}(\mu_k, I_4)$$

with identity covariance since the standardisation has already normalised the features. The number of components K is selected by BIC.

DBSCAN. Density-based clustering automatically labels low-density points as noise — precisely the anomalies we seek. The epsilon-neighbourhood radius should be calibrated to the 95th percentile of pairwise distances in clean training data.

Isolation Forest. Assigns an anomaly score based on path length to isolate a point:

$$s(z_t) = 2^{-E[h(z_t)]/c(T)}$$

where h is the isolation path length and $c(T)$ the average path length for a uniform sample of size T . Scores near 1 indicate anomalies.

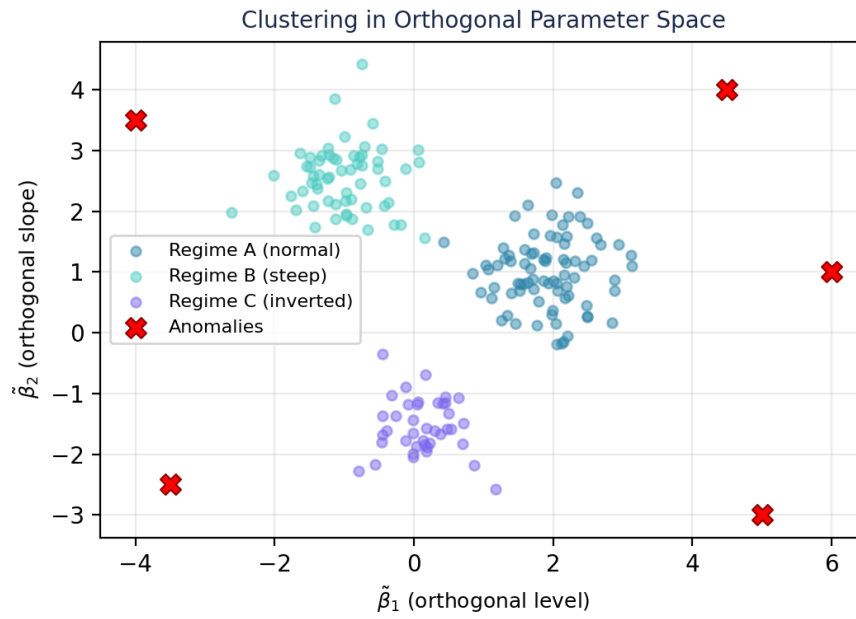


Figure 4. Clustering in a 2D projection of the orthogonal parameter space. Red crosses mark curves whose parameter signature is inconsistent with any identified regime.

6. Anomaly Detection Framework

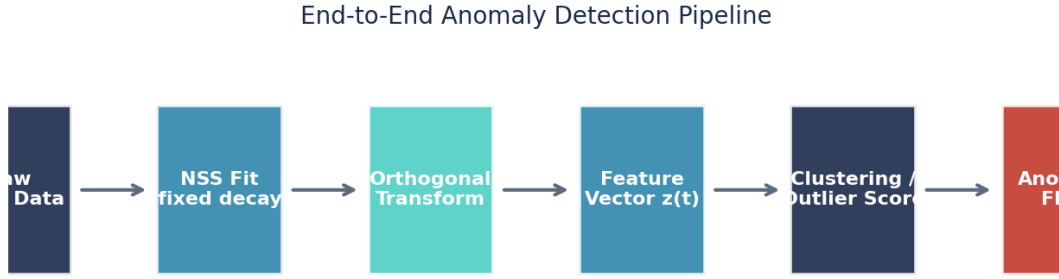


Figure 5. End-to-end pipeline from raw market data to anomaly flags.

6.1 Composite Anomaly Score

We define a composite score combining clustering and fit quality. The three terms capture distance from the nearest cluster centroid, anomalous hump asymmetry, and residual fit quality:

$$\begin{aligned}
 A_t = & w_1 \cdot \min_k \|z_t - \mu_k\| \\
 & + w_2 \cdot |\tilde{\beta}_{t,4}| / \sigma_4 \\
 & + w_3 \cdot \text{RMSE}_t / \overline{\text{RMSE}}
 \end{aligned}$$

Recommended starting weights:

$$w_1 = 0.50, \quad w_2 = 0.35, \quad w_3 = 0.15$$

The low weight on RMSE is deliberate: as argued in Section 2, RMSE is insensitive to the distortions we care about.

6.2 Statistical Threshold

Under the null hypothesis that z_t is drawn from the fitted GMM, the squared Mahalanobis distance follows a chi-squared distribution:

$$\|z_t - \mu_{k^*}\|^2 \sim \chi_4^2$$

The detection rule is:

$$\text{Flag anomaly if } \|z_t - \mu_{k^*}\|^2 > \chi_{4, 1-\alpha}^2$$

For $\alpha = 0.01$, the critical value is approximately 13.28. In practice, empirical quantiles may be preferable due to heavier-than-Gaussian tails.

6.3 Temporal Smoothing

To avoid flagging isolated noisy days, apply EWMA:

$$\bar{A}_t = \alpha A_t + (1 - \alpha) \bar{A}_{t-1}, \quad \alpha \in (0, 1)$$

A decay factor of $\alpha = 0.3$ (half-life around 2 days) works well: genuine stripping issues persist, while random noise washes out.

6.4 Regime-Conditional Thresholds

Different regimes exhibit different parameter dispersion. Compute regime-conditional thresholds:

$$\tau_k = \mu_{A|k} + \kappa \cdot \sigma_{A|k}, \quad k = 1, \dots, K$$

where the conditional moments are computed from training data assigned to cluster k , and κ is typically 2.5-3.0.

7. Practical Implementation

7.1 Decay Parameter Estimation

A two-stage approach is recommended. First, evaluate the concentrated objective on a coarse grid over plausible ranges:

$$\lambda_1 \in [0.3, 1.5], \quad \lambda_2 \in [0.05, 0.5], \quad \lambda_1 > \lambda_2$$

Then refine via constrained optimisation with minimum separation:

$$\lambda_1 - \lambda_2 \geq \delta_{min} > 0$$

A minimum separation of 0.15-0.25 ensures the two hump loadings are sufficiently differentiated.

7.2 Rolling vs. Expanding Window

Rolling window (e.g., 252 trading days): adapts to structural changes in correlation but introduces eigenvector non-stationarity. Procrustes rotation aligns eigenvectors across windows.

Expanding window with decay: more stable eigenvectors, suitable when factor structure is persistent. Half-life of 2-3 years.

7.3 Handling Sign and Ordering Ambiguity

Eigenvectors are determined only up to sign. Two remedies:

Sign alignment. Choose the sign to maximise inner product with the previous window:

$$v_j^{(t)} \leftarrow \text{sign}(v_j^{(t)T} \cdot v_j^{(t-1)}) \cdot v_j^{(t)}$$

Procrustes alignment. Find the orthogonal matrix that best aligns the full eigenvector matrix:

$$R^* = \arg \min_{R \in O(4)} \|V^{(t)} R - V^{(t-1)}\|_F$$

solved in closed form via SVD of the cross-product.

7.4 Diagnostic Checks

- **Variance concentration:** The first two factors should explain > 90% of total beta variance. If not, revisit the decay parameters.
- **Factor stability:** Track the cosine similarity between consecutive eigenvectors. Values below 0.95 warrant investigation.
- **Cluster interpretability:** Each GMM cluster should map to a recognisable market regime.
- **Backtesting:** Validate on historical episodes with known anomalies (settlement failures, auction distortions).

8. Extensions and Open Questions

8.1 Dynamic Factor Models

The orthogonal factors can be modelled as a VAR(1) to capture persistence:

$$\tilde{\beta}_t = \Gamma \tilde{\beta}_{t-1} + \eta_t, \quad \eta_t \sim \mathcal{N}(0, Q)$$

Anomalies are then large innovations:

$$A_t^{dyn} = \eta_t^T Q^{-1} \eta_t \sim \chi_4^2$$

8.2 Non-Gaussian Extensions

For heavy-tailed regimes, replace the Gaussian mixture with Student- t :

$$z_t \sim \sum_{k=1}^K \pi_k t_{\nu_k}(\mu_k, \Sigma_k)$$

This reduces false positives during crises while maintaining sensitivity to structural anomalies.

8.3 Cross-Market Consistency

For multi-currency desks, stripping errors should be cross-validated against related curves. Define:

$$A_t^{cross} = \max(A_t^{(i)} - \rho_{ij} \cdot A_t^{(j)}, 0)$$

penalising anomalies in market i that are mirrored in correlated market j (suggesting a common factor move, not a data error).

8.4 Online Adaptation

For real-time deployment, update the covariance incrementally:

$$\Sigma_{\beta}^{(t)} = (1 - \gamma) \Sigma_{\beta}^{(t-1)} + \gamma \delta \vec{\beta}_t \delta \vec{\beta}_t^T$$

where we write the centered beta as the shorthand. Eigenvectors can be tracked via incremental SVD methods, avoiding recomputation of the full decomposition each day.

8.5 Anomaly Classification

The decomposition not only flags anomalies but classifies them:

Anomalous Factor	Likely Cause	Recommended Action
Level only	Data feed lag / stale quotes	Wait and re-observe
Slope only	Short-end quote error	Check money market inputs

Curvature (belly)	Stripping artefact at 5-7y	Inspect strip vs. bond prices
Hump asymmetry	Spurious 2nd hump; interpolation failure	Review raw bond set; check missing bonds
Multiple factors simultaneously	Genuine market event or systemic data issue	Cross-validate with other markets

Table 2. Anomaly classification based on which orthogonal factor exceeds its threshold.

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